Amazon Product Sales Analysis

Drivers of Revenue, Quality, and Competitive Positioning

Field	Details
Sector	E-Commerce / Retail Technology
Team ID	DETECTIVES
Team Members	Vipul Yadav · Yash Kishor Mali · Ananya Gupta · Yashveer Singh · Abhay Pratap Yadav
Faculty Mentor	Vrushali
Date	28 April 2026

Context and Problem Statement

Sector Context

- Amazon holds **>38% of US e-commerce market share**
- Sellers compete on algorithmic discoverability: ratings, reviews, badge signals, Buy Box ownership
- Price alone is insufficient — social proof and promotional levers now drive purchase velocity

Core Business Question

> *Which product attributes most strongly drive estimated monthly revenue — and which levers should sellers prioritise?*

Objectives

1. Identify top revenue drivers through correlation, regression, and hypothesis testing
2. Quantify revenue gaps between quality tiers and badge types
3. Surface actionable, category-specific recommendations via a Tableau dashboard

Data Engineering

Metric	Value
Raw rows	42,675
Cleaned rows	30,228
Retention rate	70.8%
Categories covered	15+ electronics sub-categories
Scrape date	2025-08-21 (single snapshot)

Major Cleaning Steps

1. Dropped duplicates; coerced numeric columns from string to float
2. Dropped rows missing core fields (`discounted_price`, `product_rating`, `purchased_last_month`)
3. Validated: price > 0, rating 0–5, discount ≤ 90%
4. Engineered `estimated_revenue`, `discount_value`, `rating_category`

KPI Framework

KPI	Formula	Business Purpose
Estimated Revenue	`purchased_last_month × discounted_price`	Primary revenue proxy per product
Total Category Revenue	`Σ estimated_revenue` per category	Category investment priority
Average Rating	`mean(product_rating)`	Portfolio quality health
Review Volume	`mean(total_reviews)`	Social proof and discoverability
Best Seller Penetration	Best seller listings / total	Badge capture rate
Buy Box Win Rate	Buy Box available / total	Listing competitiveness
Coupon Adoption Rate	Coupon listings / total	Promotional activity level
Sponsorship Rate	Sponsored listings / total	Paid visibility investment

Key EDA Insights

Insight	Value
Total estimated revenue (30,228 products)	**$1.575 Billion**
Average product rating	**4.44** (median 4.5) — compressed range
Products rated ≥ 4.0	**91.3%** — quality is a baseline requirement
High-rated products' revenue share	**95.8%** ($1.508B of $1.575B)
Avg monthly purchases per product	**1,354 units**
Best Seller badge holders	**0.9%** of listings

Top Categories by Total Revenue

Rank	Category	Total Revenue	Mean Rev / Listing
1	Power & Batteries	$526.7M	$196,303
2	Laptops	$312.4M	$51,740
3	Phones	$240.1M	—

> Top 3 by **mean** revenue: Power & Batteries ($196k), Laptops ($52k), Smart Home ($50k)

Statistical Testing — Results

n = 30,228 · α = 0.05 · Source: `notebooks/04_statistical_analysis.ipynb`

Test	Result	p-value	Decision
Pearson r — `total_reviews` / revenue	r = **0.208**	< 0.001	Reject H0 — reviews predict revenue
Pearson r — `discount_%` / revenue	r = **0.000**	0.994	Fail to reject H0 — **no effect**
Welch's T-Test — Best Seller badge	$510k vs $48k (10.6×), t = **7.17**	7.74×10^{-12}	Reject H0 — badge = significant premium
One-Way ANOVA — 15 categories	F = **161.2**	< 0.001	Reject H0 — categories differ
Chi-Square — coupon × badge	χ^2 = **0.480**	0.489	Fail to reject H0 — **independent**

OLS Model (R^2 = 0.180, F = 944.9)

Predictor	Coefficient	p-value
Best Seller badge	**+$396,500/mo**	< 0.001
Sponsored	**+$156,700/mo**	< 0.001
Coupon	**−$56,470/mo**	< 0.001
Reviews	**+$1.74/review**	< 0.001

Dashboard Overview

Built in Tableau · Data: `data/processed/final_tableau_data.csv`

Executive View

- KPI tiles: Total revenue · Avg rating · Avg discount % · Buy Box win rate
- Ranked bar chart: Category revenue by listing count
- Rating histogram: Portfolio quality distribution

Operational View

- Sortable product table: Revenue, rating, reviews, discount %, badge status
- Scatter plot: `total_reviews` vs `estimated_revenue` by `rating_category`
- Box plot: Revenue distribution by category

Filters

- Category multi-select · Rating range slider
- Badge filter (Best Seller / No Badge / Limited Time Deal)
- Buy Box toggle · Sponsored / Organic toggle

Top Insights

1 — Best Seller Badge = 10.6× Revenue Premium

Badge holders: **$510,036/mo** vs **$48,141/mo** non-badge ($t = 7.17$, $p = 7.74 \times 10^{-12}$) · OLS: **+$396,500/mo**

2 — Sponsored Placement is the Strongest Controllable Lever

OLS coefficient = **+$156,700/month** ($p < 0.001$) — achievable without earning the badge first

3 — Discounts Don't Drive Revenue — Coupons Actively Hurt It

Discount %: $r =$ **0.000** ($p = 0.994$) — zero relationship

Coupons: OLS **–$56,470/mo** ($p < 0.001$) — revenue-negative once controlled

4 — Reviews: High ROI, Near-Zero Cost

OLS: **+$1.74 per additional review** ($p < 0.001$)

10 SKUs from 200 → 1,000 reviews ≈ **$13,920/month** incremental

Recommendations

> Full evidence chains in `docs/recommendations.md`

Priority	Recommendation	Estimated Impact
High	**Earn the Best-Seller badge**	5 SKUs ≈ **$1.15M/month**
High	**Stop discounting — shift to sponsored**	≈ **$1.4M/month** margin recovery
Medium	**Cut coupons → sponsored campaigns**	Up to **$10M/month** across 50 SKUs
Medium	**Concentrate spend on top categories**	**15–25% uplift** on reallocated budget
Medium	**Drive reviews on low-review SKUs**	10 SKUs ≈ **$13,920/month**

Impact and Feasibility

Action	Timeframe	Estimated Impact
Best Seller badge push (top 5 SKUs)	60-day pilot → 6–12 months	≈ $1.15M/month
Discount reduction	30-day A/B test	≈ $1.4M/month margin recovery
Coupon → sponsored shift (50 SKUs)	30-day pilot	Up to $10M/month
Category reallocation (top 3)	2-week ramp	15–25% uplift
Review-velocity campaign (10 SKUs)	90 days	≈ $42k

Recommendation	Feasibility	Revenue Impact
Best Seller badge targeting	Medium	Very High
Stop discounting / margin recovery	High	High
Coupon → sponsored shift	High	Very High
Category spend concentration	Medium	Very High
Review velocity campaign	High	Medium

Limitations

Data Constraints

- Single snapshot (2025-08-21) — no seasonality analysis possible
- Revenue is a proxy — rounding in purchase counts; excludes returns, fees
- No COGS or advertising spend data — profit analysis not possible
- Sustainability tags 92.1% null — not usable as predictor

Method Constraints

- OLS $R^2 = 0.180$ — 82% of variance unexplained (product age, ad history, seasonal demand)
- Reverse causation risk: sellers sponsor and solicit reviews for already-performing SKUs
- Large n (30,228): small effects appear statistically significant — check practical significance
- H5 corrected: chi-square $p = 0.489$ — coupons and badge status are **independent**

Contribution Matrix

This project was completed as a collaborative effort across all stages of the data pipeline, analysis, and dashboard development.

Team Member	Role
Vipul Yadav	Data cleaning, preprocessing, feature engineering, Tableau dashboard publishing
Yash Kishor Mali	Project documentation, README, repository structure
Ananya Gupta	Dashboard design, KPI creation, Tableau visualisations
Yashveer Singh	Statistical analysis, hypothesis testing, key insights and recommendations
Abhay Pratap Yadav	Data extraction, project report, presentation

Future Extensions & Closing

Next Steps

Extension	Description
Time-series collection	6–12 months for seasonality modelling
Log-linear regression	Better fit for right-skewed revenue
K-means clustering	Segment products into natural archetypes
NLP on product titles	Keyword and sustainability tag effects
Profit margin modelling	Integrate advertising cost and COGS

Closing Summary

> Amazon success is **not** won on price or discounts ($r = 0.000$, $p = 0.994$). It is won through the **Best Seller badge** (+$396k/mo), **sponsored placement** (+$157k/mo), **review volume** (+$1.74/review), and **right category** (ANOVA $F = 161.2$).

Thank You

Questions Welcome

Vipul Yadav · Yash Kishor Mali · Ananya Gupta · Yashveer Singh · Abhay Pratap Yadav
DVA Capstone · Newton School Of Technology · 28 April 2026